ELEC70022
EE9-CS6-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday, 2 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : D. Angeli

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MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. Consider a compound pendulum and an Earth-fixed Cartesian coordinate system with its origin at point O and with unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ (not shown), and $\boldsymbol{k}$, as shown in Figure 1.1. The unit vector $\boldsymbol{j}$ is horizontal and normal to the page. The pendulum rotates only about the axis through O in the $\boldsymbol{j}$ direction by an angle θ under the influence of gravity. The axis of rotation is at an offset l from the centre of mass O_{cm}. The mass of the compound pendulum is m and its inertia matrix with respect to body-fixed axes with origin the centre of mass is

$$\boldsymbol{I}_{\text{cm}} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

where I_{yy}, I_{zz}, and I_{xx} are the moments of inertia about the horizontal spin axis, the axis joining points O and O_{cm}, and the axis perpendicular to the two previously mentioned axes, respectively., and I_{xy}, I_{yx}, I_{xz}, I_{zx}, I_{yz}, I_{zy}, are the corresponding products of inertia.

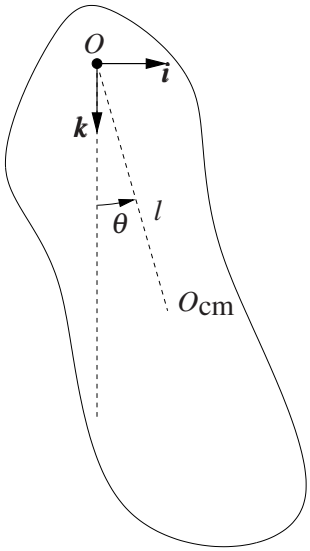


Figure 1.1 Compound pendulum.

Use body-fixed coordinates to answer the questions below.

a) Calculate the inertia matrix of the compound pendulum with respect to body-fixed axes that are parallel to the ones used to define $\boldsymbol{I}_{\text{cm}}$ and with origin the point O. [3]

b) Write the angular velocity vector of the compound pendulum. [2]

c) Compute the angular momentum vector of the compound pendulum. [3]

d) Use the vectorial approach to derive:

   i) the equation of motion of the pendulum, and [4]

   ii) the constraint moments acting from Earth on the compound pendulum at the joint O that prevent the compound pendulum from rotating about any axes other than in the $\boldsymbol{j}$ direction. [8]

2. As shown in Figure 2.1, a uniform solid object rotates uniformly with a constant angular speed of ω_1 about a vertical axis passing through the fixed centre of mass of the object, indicated by point C in Figure 2.1. The solid object is axisymmetric with a spin (about the axis of symmetry) moment of inertia I_{zz} and a radial moment of inertia, passing through the centre of mass, I_{xx}. The vertical axis of rotation is at an angle θ with the axis of symmetry, as shown in Figure 2.1. Angle θ is increasing at a constant rate of $\dot{\theta} = \omega_2$.

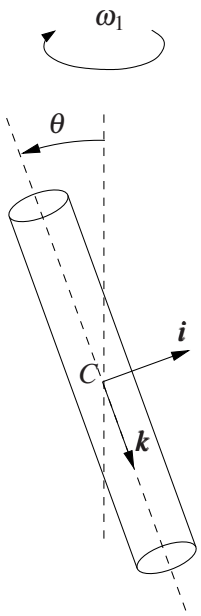


Figure 2.1 Axisymmetric solid object.

An unspun (with respect to the solid object) Cartesian coordinate system with its origin at point C and unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ (not shown) and $\boldsymbol{k}$, with $\boldsymbol{k}$ along the axis of symmetry of the object, as shown in Figure 2.1, is used to analyse the motion of the solid object. The coordinate system rotates with the solid object by the angular speed ω_1 and ω_2 but it does not spin (about the axis of symmetry) with the solid object.

a) By using the unspun coordinate system, compute the kinetic energy and potential energy of the solid object, and hence determine the Lagrangian function. [4]

b) By considering the rotational motion about the fixed point C, use the Lagrangian approach to calculate the component in the $\boldsymbol{j}$ direction of the moment required to maintain the motion of the solid object. [5]

c) Assume the solid object spins about its axis of symmetry with a constant speed of ω_3 in addition to ω_1 and ω_2.

   i) By using the unspun coordinate system, compute the kinetic energy and potential energy of the solid object, and hence determine the Lagrangian function. [4]

   ii) Compute the additional moment in the $\boldsymbol{j}$ direction, as compared to the moment found in part c), that is required to maintain the motion of the solid object. [7]

3. As in Q2 and as shown in Figure 3.1, a uniform solid object rotates uniformly with a constant angular speed of ω_1 about a vertical axis passing through the fixed centre of mass of the object, indicated by point C in Figure 3.1. The solid object is axisymmetric with a spin (about the axis of symmetry) moment of inertia I_{zz} and a radial moment of inertia, passing through the centre of mass, I_{xx}. The vertical axis of rotation is at an angle θ with the axis of symmetry, as shown in Figure 3.1. Angle θ is increasing at a constant rate of $\dot{\theta} = \omega_2$.

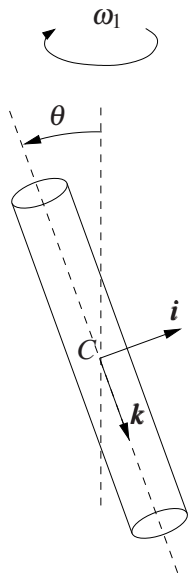


Figure 3.1 Axisymmetric solid object.

An unspun (with respect to the solid object) Cartesian coordinate system with its origin at point C and unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ (not shown) and $\boldsymbol{k}$, with $\boldsymbol{k}$ along the axis of symmetry of the object, as shown in Figure 3.1, is used to analyse the motion of the solid object. The coordinate system rotates with the solid object by the angular speed ω_1 and ω_2 but it does not spin (about the axis of symmetry) with the solid object.

a) Calculate the angular velocity vector of the solid object in the unspun coordinate system. [2]

b) Compute the angular momentum vector of the solid object in the same unspun coordinate system. [3]

c) By considering the rotational motion about the fixed point C, use the vectorial approach in unspun coordinates to calculate the moment vector required to maintain the motion of the solid object. [5]

d) Assume the solid object spins about its axis of symmetry with a constant speed of ω_3 in addition to ω_1 and ω_2.

   i) Write the angular velocity vector of the solid object in the unspun coordinate system. [2]

   ii) Determine the angular momentum vector of the solid object in the unspun coordinate system. [2]

   iii) Compute the additional moment, as compared to the moment found in part c), that is required to maintain the motion of the solid object. [6]

4. As in Q1, consider a compound pendulum and an Earth-fixed Cartesian coordinate system with its origin at point O and with unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ (not shown), and $\boldsymbol{k}$, as shown in Figure 4.1. The unit vector $\boldsymbol{j}$ is horizontal and normal to the page. The pendulum rotates only about the axis through O in the $\boldsymbol{j}$ direction by an angle θ under the influence of gravity. The axis of rotation is at an offset l from the centre of mass O_{cm}. The mass of the compound pendulum is m and its inertia matrix with respect to body-fixed axes with origin the centre of mass is

$$\boldsymbol{I}_{\text{cm}} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

where I_{yy}, I_{zz}, and I_{xx} are the moments of inertia about the horizontal spin axis, the axis joining points O and O_{cm}, and the axis perpendicular to the two previously mentioned axes, respectively., and I_{xy}, I_{yx}, I_{xz}, I_{zx}, I_{yz}, I_{zy}, are the corresponding products of inertia.

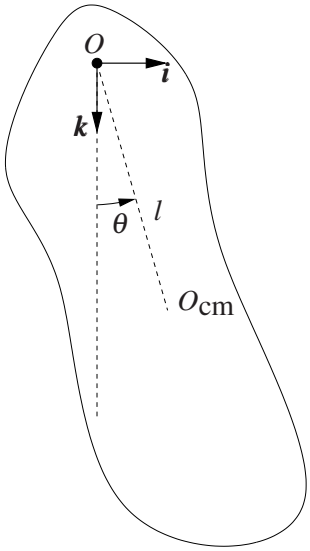


Figure 4.1 Compound pendulum.

a) By introducing a second rotational degree of freedom in the motion of the compound pendulum about the fixed point O and a corresponding kinematic constraint, use the Lagrangian approach to calculate the constraint moment acting from Earth on the compound pendulum that prevents the pendulum from rotating about the axis joining points O and O_{cm}. To achieve this, complete the following steps:

   i) Define appropriate pendulum rotational degrees of freedom about point O with the associated generalised coordinates, an Euler angle convention, and an appropriate kinematic constraint. [3]

   ii) Write the angular velocity vector of the compound pendulum using a body-fixed coordinate system with its origin at point O. [2]

   iii) Calculate the kinetic energy, potential energy and the Lagrangian function of the compound pendulum. [2]

   iv) Compute the constraint moment. [3]

b) By introducing a second rotational degree of freedom in the motion of the compound pendulum about the fixed point O and a corresponding kinematic constraint, use the Lagrangian approach to calculate the constraint moment acting from Earth on the compound pendulum that prevents the pendulum from rotating about the axis that is perpendicular to both the axis joining points O and O_{cm}, and the axis through O in the $\boldsymbol{j}$ direction. To achieve this, complete the following steps:

i) Define appropriate pendulum rotational degrees of freedom about point O with the associated generalised coordinates, an Euler angle convention, and an appropriate kinematic constraint. [2]

ii) Write the angular velocity vector of the compound pendulum using a body-fixed coordinate system with its origin at point O. [2]

iii) Calculate the transformation matrix from these body-fixed to Earth-fixed coordinates. [2]

iv) Compute the kinetic energy, potential energy and the Lagrangian function of the compound pendulum. [2]

v) Calculate the constraint moment. [2]